

Sample Project Documentation Guide

Date	06 July 2026
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Team

Comprehensive Project Implementation Manual

Project Title: Smart Lender - Credit Risk evaluation platform

Abstract: Financial institutions experience inefficiencies and high turnaround times when auditing applicant loan eligibility. Smart Lender addresses this gap by implementing an automated machine learning classification model built on Flask. The model preprocesses categorical columns using mode imputation and numerical columns using median imputation. It balances target class distributions using SMOTE and applies StandardScaler scaling, achieving high testing accuracy. Real-time evaluation yields predictions along with confidence levels and identifies critical risk factors (low income, poor credit score, high DTI ratios).

Milestone Breakdown Summary

Milestone 1	Model Selection & Architecture: Preprocess and train KNN, Decision Tree, Random Forest, and XGBoost. Best model (XGBoost) selected.
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